

LUND UNIVERSITY

Mathematics, Faculty of Science

Number theory
MATM35
HT 2020
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Written examination (online)
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1. Determine, using Legendre symbols, whether the congruences

$$x^2 + 3x + 5 \equiv 0 \pmod{607} \quad \text{and} \quad x^2 + 3x + 5 \equiv 0 \pmod{647},$$

respectively, are solvable or not. (Note that 607 and 647 are primes.)

2. Given the continued fraction expansion $\sqrt{3} = [1; \overline{1, 2}]$, find a convergent p_n/q_n that satisfies

$$\left| \sqrt{3} - \frac{p_n}{q_n} \right| \leq \frac{1}{1000}.$$

3. For any positive integer n with prime factorization $n = p_1^{k_1} p_2^{k_2} \cdots p_r^{k_r}$, prove that

$$\sum_{d|n} \mu(d)\sigma(d) = (-1)^r p_1 p_2 \cdots p_r,$$

where μ is the Möbius function and σ is the sum-of-divisors function.

4. Let p and q be primes and suppose that p is of the form $4k + 3$ and $q = 2p + 1$. Prove that q divides $2^p - 1$.

5. Let p be a prime and define

$$S_n = \sum_{j=1}^{p} j^n$$

for any positive integer n. Prove that either $S_n \equiv 0 \pmod{p}$ or $S_n \equiv -1 \pmod{p}$. For which n does the second alternative hold?

6. If p is an odd prime, then either $p = 4n + 1$ or $p = 4n - 1$ for a positive integer n. In both these cases:

   (a) Prove that n is a quadratic residue $\pmod{p}$,

   (b) Compute $n^n \pmod{p}$.

   (In (b), the answer should be between $-p/2$ and $p/2$.)

Give careful motivations for your solutions!